

The WealthLine—Examining the Effect of the HealthLine Bus Rapid Transit on Resident Employment

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Abstract

This paper analyzes the impact of the HealthLine Bus Rapid Transit (BRT) line, opened in 2008 to connect two major economic hubs in Cleveland, on local resident employment outcomes. With large federal grants allocated towards public transit nationwide, understanding the effects of transit expansion becomes increasingly important. Using the Longitudinal Employer-Household Dynamics Origin-Destination Employment Statistics (LODES) dataset and Census Block Group level American Community Survey (ACS) 5-year data from 2004-2013, we utilize two-way fixed effects difference-in-differences and triple difference-in-differences methodologies, to analyze the impact of the HealthLine opening on resident employment at the Census Block Group level in Cuyahoga County. We find that while the HealthLine's introduction did not significantly affect overall employment for residents living along the Euclid corridor, it did raise the healthcare job share for these communities by 3 percentage points. Notably, for areas with higher educational attainment, we found significant increases to overall employment, showing the importance of increased education in conjunction with transit access. These findings inform urban development and revitalization strategies that integrate increased education access with public transit improvements.

Keywords: Bus Rapid Transit, Employment, Cleveland, HealthLine, Employment Outcomes, Economic Development, Public Transportation, Urban Development, Urban Infrastructure

1 Introduction

In October 2008, Cleveland’s Rapid Transit Authority (RTA) opened its first Bus Rapid Transit (BRT) line along the Euclid Corridor.¹ The goal of the project was to revitalize the former Millionaire’s Row by connecting Cleveland’s two major economic hubs—Downtown and University Circle.² Residents on the line gained more reliable, direct access to major Cleveland institutions like the Cleveland Clinic, University Hospitals, Case Western Reserve University, Cleveland State University, Severance Hall, and the Cleveland Museum of Art via public transportation.³ Existing literature has established that public transportation access is associated with increased employment. Furthermore, the HealthLine BRT has been credited with sparking investments in the Health-Tech Corridor of Cleveland.⁴ The HealthLine name was the result of a \$6.25 million naming rights deal with the Cleveland Clinic and University Hospitals.⁵ In 2018, the RTA estimated that the establishment of the HealthLine led to around \$9.5 billion of investments in the Euclid Corridor.⁶ This was an overall return of \$190 for each dollar spent on developing the HealthLine project.⁷ Overall, the Cleveland HealthLine clearly prompted economic revitalization of the region in the form of investments. However, little is known about the economic impact on the residents around the HealthLine.

In recent years, public transportation development has become a larger issue in the United States. Senator Sherrod Brown recently announced \$66.3 million in federal funding, as part of the Infrastructure Investment and Jobs Act, for the development of public transportation in Cleveland.⁸ This year, the Cleveland RTA was awarded a \$700,000 grant for research into Bus Rapid Transit development in Cleveland’s Lorain Corridor.⁹ With this increased attention towards public transit investment, understanding how these investments affect resident communities is paramount. We therefore aim to investigate whether the increased opportunities created by the HealthLine project went to those living around the HealthLine. In the absence of detailed person-level data, we use Census Block Group level data, investigating the area as a whole. Our research aims to shed light on the impact of public transit projects on the marginalized communities that often surround these lines.

¹Oct. 24, 2008: *Healthline opens, pumping new life into Cleveland*, Oct. 2008, <https://www.riderta.com/news/oct-24-2008-healthline-opens-pumping-new-life-cleveland>.

²*Euclid Avenue Healthline Bus Rapid Transit*, <https://www.sasaki.com/projects/euclid-avenue-healthline-bus-rapid-transit/>.

³Oct. 24, 2008: *Healthline opens, pumping new life into Cleveland*, Oct. 2008, <https://www.riderta.com/news/oct-24-2008-healthline-opens-pumping-new-life-cleveland>.

⁴Jeff Epstein, “Cleveland Health-tech corridor”, *Economic Development Journal* 15, no. 2 (2016): 7–13, ISSN: 1539-1922, <https://web.p.ebscohost.com/ehost/pdfviewer/pdfviewer?vid=2&sid=80851b1c-cbfe-4b7c-b5f0-2f2906505ecd@redis>.

⁵*Clinic, UH pay to name Euclid Corridor buses - cleveland.com*, https://www.cleveland.com/medical/2008/02/clinic_uh_pay_to_name.html.

⁶Grant Segall, *RTA says HealthLine had 10-year payback of \$9.5 billion, ‘woke up’ Euclid Corridor*, news, Nov. 2018, <https://www.cleveland.com/news/erry-2018/11/149927818e3851/rta-says-healthline-had-10year.html>.

⁷*Ibid.*

⁸*Brown Announces \$264 Million for Ohio’s Public Transit Systems — U.S. Senator Sherrod Brown of Ohio*, <https://www.brown.senate.gov/newsroom/press/release/sherrod-brown-announces-264-million-for-ohios-public-transit-systems-fy2023>.

⁹Sam Allard, *RTA gets federal grant for Lorain Corridor*, Apr. 2024, <https://www.axios.com/local/cleveland/2024/04/09/rta-federal-grant-lorain-corridor-brt>.

2 Literature Review

There is extensive literature regarding public transportation and its links to employment and economic output. There have been many studies looking at the effect of new transit lines in various cities around the world. In 2006, Yi studied the impact of public transportation on employment status in Houston, Texas.¹⁰ Yi found that public transportation access has significant positive effects on employment status, despite varying levels of effect on various socio-economic classes.¹¹ Our study aims to localize these findings to Cleveland residents living around the HealthLine. We look to observe the impact of business district BRT, a relatively low-cost infrastructure solution, on employment for local residents.

Piiparinen and Russell write about the role of public transportation in the economic restructuring of Cleveland.¹² They find that Cleveland has seen an increase in healthcare jobs, and overall higher paying jobs in the Health-Tech Corridor.¹³ The Cleveland HealthLine runs through the Health-Tech Corridor, connecting Downtown Cleveland with the major hospitals in University Circle. Epstein credits the Cleveland HealthLine with spurring the rapid development of the Health-Tech Corridor.¹⁴ This increase in job quality around and because of the HealthLine prompts our research into the benefits felt by residents surrounding this increased economic activity.

Nelson et al. conducted a micro analysis of the BRT system in Eugene-Springfield Oregon on job creation along the new line.¹⁵ They used workplace area characteristics data from the Longitudinal Employment-Household Dynamics (LEHD) Origin-Destination Employment Statistics (LODES) dataset. They found a positive correlation between distance to BRT stations and job creation. In 2016, Nelson, Kim, and Ganning looked at post-Great Recession growth around multiple BRT lines around the country including the HealthLine.¹⁶ They focused on workplace area characteristics, and the addition of jobs in Census Blocks surrounding these BRT lines using the same LODES dataset.¹⁷ We aim to use the residents portion of this dataset to observe not whether jobs are created around the HealthLine, but whether residents around the HealthLine received an employment boost as a result of the new lines. Credit's dissertation looks at business

¹⁰Chang Yi, "Impact of Public Transit on Employment Status: Disaggregate Analysis of Houston, Texas", *Transportation Research Record Journal of the Transportation Research Board* 1986, no. 1 (Jan. 2006): 137–144, <https://doi.org/http://dx.doi.org/10.1177/0361198106198600117>.

¹¹Ibid.

¹²Richey Piiparinen and Jim Russell, "Center for Population Dynamics Quarterly Brief January 2017: Transportation's Role in the Economic Restructuring of Cleveland", *All Maxine Goodman Levin School of Urban Affairs Publications*, Jan. 2017, https://engagedscholarship.csuohio.edu/urban_facpub/1427/.

¹³Ibid.

¹⁴Jeff Epstein, "Cleveland Health-tech corridor", *Economic Development Journal* 15, no. 2 (2016): 7–13, ISSN: 1539-1922, <https://web.p.ebscohost.com/ehost/pdfviewer/pdfviewer?vid=2&sid=80851b1c-cbfe-4b7c-b5f0-2f2906505ecd@redis>.

¹⁵Arthur C. Nelson et al., "Bus Rapid Transit and Economic Development: Case Study of the Eugene-Springfield BRT System", *Journal of Public Transportation* 16, no. 3 (July 2013): 41–57, <https://doi.org/http://doi.org/10.5038/2375-0901.16.3.3>.

¹⁶Arthur C. Nelson, Keuntae Kim, and Joanna P. Ganning, "Bus Rapid Transit and Economic Development: A Quasi-Experimental Treatment and Control Analysis" (Washington DC, United States: University of Arizona: College of Architecture, Planning, / Landscape Architecture, University of Utah: Department of City / Metropolitan Planning, Cleveland State University, Nov. 2016), 23, https://ppms.trec.pdx.edu/media/project_files/Nelson__Kim_-_Bus_Rapid_Transit_and_Economic_Development.pdf.

¹⁷Ibid.

creation in proximity to public transportation.¹⁸ By examining the relationship between transit proximity and new startups, he finds a statistically significant relationship between number of businesses and transit station proximity. Nelson, Kim, and Ganning confirm this effect, finding that HealthLine-adjacent blocks saw less business attrition during the recession, supporting the idea that the HealthLine promotes job creation. Our paper intends to measure whether the jobs that are created go to those who live in direct proximity to the line.

More recently, Pasha et al. looked at overall public transit access in Cuyahoga County, and its effect on low income persons, especially in regards to displacement.¹⁹ They find that public transportation increases access to jobs, while not increasing property values, indicating that public transportation can be an effective tool at improving outcomes for low-income individuals.²⁰ Our paper builds existing literature by looking at job growth for residents and using a difference-in-differences model instead of a synthetic placebo model like in the Nelson, Kim, and Ganning paper. Nelson et al. also focuses specifically on post-recession data, which in the context of the Cleveland HealthLine, focuses around the opening of the line, without much data from before the line opening. Our paper looks to investigate the effect of increased public transportation access on job opportunities by examining the addition of a major flagship transit line in Cleveland.

3 Research Method

3.1 Estimation Strategy

Our analysis begins with multivariate regressions (including interaction terms and Fixed Effects for year and block group) and then considers a Difference-in-Differences estimator, utilizing Two-Way Fixed Effects and a Triple Difference with interaction terms. The variable of interest—HealthLine access—had to be defined using GIS software.²¹

3.1.1 Defining HealthLine Access

HealthLine access was defined as any Census Block Group (according to the 2020 geographic definitions) that has over 30% of its total area within 800 meters of the HealthLine. Census Block Group definitions were sourced from the U.S. Census Bureau, and the HealthLine geographic definition was manually collected using its stop locations and collecting coordinates in a kml file. The map in Figure 1 shows the HealthLine buffer that was created as well as locations for major Cleveland Institutions (namely Cleveland State University, the Cleveland Clinic, University Hospitals, and Case Western Reserve University)

¹⁸Kevin N. Credit, “Economic development for the 21st century: how proximity to transit and walkability influence business creation and performance” (PhD thesis, Michigan State University, 2018), <https://doi.org/doi:10.25335/bvec-qy72>.

¹⁹Obed Pasha et al., “Transit effects on poverty, employment, and rent in Cuyahoga County, Ohio”, *Transport Policy* 88 (Mar. 2020): 33–41, <https://doi.org/https://doi.org/10.1016/j.tranpol.2020.01.013>.

²⁰Ibid.

²¹QGIS, an open source GIS software, was used to process manually collected geographic data, a log of the processing can be found on the github repository: https://github.com/jmccormick7/HealthLineEmploymentStudy/blob/main/QGIS_history.log

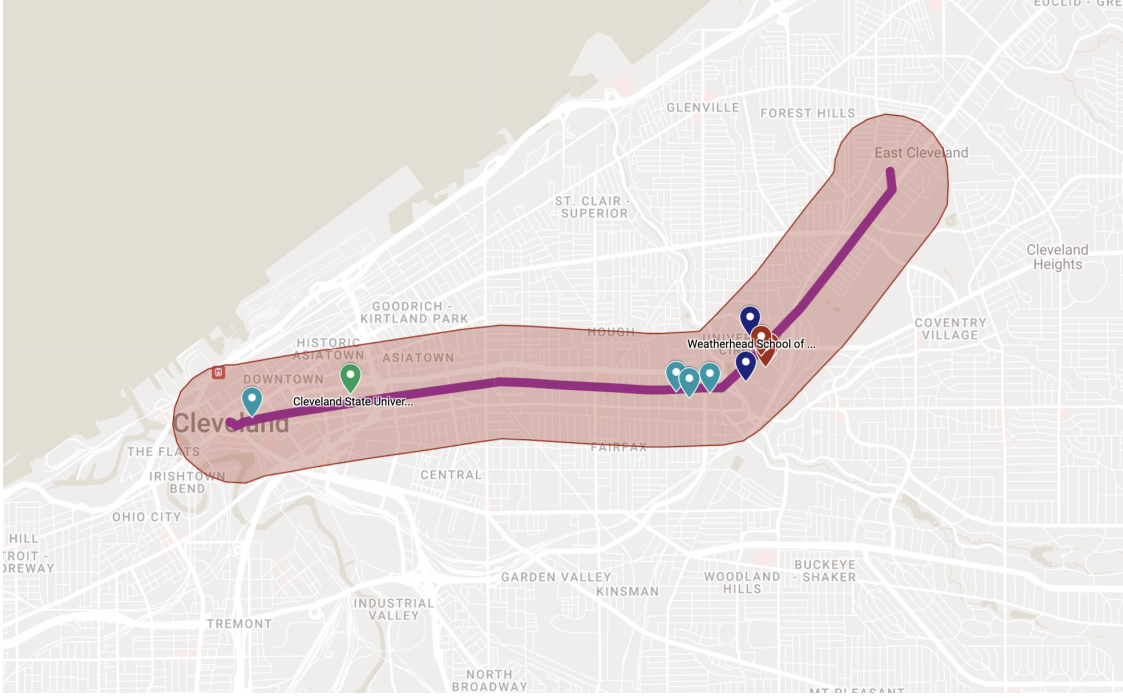


Figure 1: 800m buffer around the Cleveland HealthLine BRT. Block Groups with over 30% of their area within this purple buffer are defined as having access to the HealthLine (Treatment) while the rest do not (Control).

3.1.2 Difference-in-Differences Estimator

The primary model considers the introduction of the HealthLine in late 2008. The years of observation are then split into two groups, a before opening group (2004-2008) and an after opening group (2009-2013). This regression is run for multiple outcome variables. A Difference-in-Differences approach is used due to our treatment being introduced only to a certain region at this opening time, creating an opportunity for this framework to be applied. The outcome variables are listed below:

1. *% Pop Employed* – Percent of Total Population Employed
2. *Health Care Social Assistance Job Share* – Job Share in Health Care/Social Assistance Industry
3. *Educational Services Job Share* – Job Share in Educational Services Industry
4. *Prof/Sci/Tech Job Share* – Job Share in Professional/Scientific/Technical Services Industry
5. *Accommodation and Food Services Job Share* – Job Share in Accommodations/Food Services Industry

The model is as follows:

$$Y_{itj} = \beta_0 + \beta_1 \text{HealthLine}_{it} + \beta_2 \text{After}_{it} + \beta_3 \text{HealthLine} \times \text{After}_{it} + \beta_\mu \mu_{it} + \epsilon_{it}$$

Where:

i : Census Block Group (2020 Census Designation)

j : Outcome variable

t : Year (2004-2013)

Y : Outcome variable (Listed above)

μ : Covariate Controls (listed below)

ϵ : Error

The covariate controls that are used are defined below:

1. *% Black* – Percent of Census Block Group population that is Black
2. *% No GED* – Percent of Census Block population that has not obtained a GED
3. *% Has Degree* – Percent of Census Block Group population that has at least an Associate's Degree
4. *Med Home Val* – Median Home Value of the Census Block Group

3.2 Identification Strategy

The data ideally would consist of yearly demographic controls, with detailed reported income, and working populations. The ideal data would also contain information about HealthLine or public transit ridership and tie that to individual income and employment records. In lieu of this ideal data, public transit ridership/efficacy is defined by the availability of public transit. That is whether residents live in areas that can reasonably access the HealthLine (over 30% of the Census Block Group area being within 800m of the HealthLine). Broad income levels are available through the LODES dataset, with bins of under \$1250 per month, between \$1250 and \$3333 per month, and over \$3334 per month. That is annual incomes of under \$15,000, between \$15,000 and \$39,996, and over \$39,996 respectively.

Furthermore, the U.S. Census Bureau does not collect detailed demographic data at the Census Block nor the Census Block Group level at a yearly rate. In lieu, of this ideal demographic control data, 5-year ACS summary data at the Census Block Group level is used. with the 2009 5-year ACS covering the pre-treatment period and the 2013 5-year ACS covering the post-treatment period.

The data used does have unobserved heterogeneity in that individual job seekers are unknown. The HealthLine treatment area is, on average, over 20 percentage points less white than the control area. Using the percentage of White residents in a Census Block Group aims to control the economic disparities between communities of color and White communities. Furthermore, holding constant education level (the percentage of the Block Group population that holds at least an Associates Degree) aims to control for differences in overall skill between areas. Median home values of Census Block Groups (in 2013 U.S Dollars) is used to control for neighborhood effects like school district quality, crime, and all other factors that impact both property value and employ-ability.

There is no issue with reverse causality as there was no major demographic shifts in communities surrounding the HealthLine, suggesting that there was not a displacement effect in which the HealthLine led those already employed living in control areas to displace unemployed residents in the treatment areas. While there certainly could have been instances of this, on a whole this seems to be unlikely due to the demographic similarities of the treatment group pre-treatment and post-treatment.

There is a potential measurement error that comes up because of the broadness of population data. The population data is sourced from the 5-year ACS, thus it is a 5-year average of the population that the yearly employment data is normalized on. While this could cause problems, ultimately the likelihood is that population and demographics would not vary significantly enough to alter the data used for estimation. And any demographic changes, or population changes caused by the opening of the HealthLine would be caught by the difference between the pre-treatment average and the post-treatment average.

4 Data

The data for this paper is sourced from the U.S. Census Bureau. Through both the U.S. Census Bureau itself, and IPUMS NHGIS. Employment data comes from the Longitudinal Employer-Household Dynamics (LEHD) Origin-Destination Employment Statistics (LODES) dataset²², specifically the Residence Area Characteristics (RAC) data. Covariate controls were sourced from 5-year ACS summary data at the Census Block Group level through IPUMS NHGIS. The 2009 5-year ACS data²³ and the 2013 5-year ACS data²⁴ were used for pre-treatment and post-treatment controls respectively. The geographic crosswalk files that were used to convert the ACS Census Block Group designations to the 2020 definitions were sourced from IPUMS NHGIS. A 2000 to 2010 crosswalk²⁵ and a 2010 to 2020 crosswalk²⁶ were utilized. These crosswalk files contained geographic population weights that were used to take weighted averages of population when converting, to account for block groups being split up into multiple block groups. A U.S. Census Blocks shapefile from the U.S. Census Bureau was obtained from Cuyahoga County Open Data.²⁷ And the HealthLine geographic data was manually obtained using mapping software and QGIS. The U.S. Census Blocks shapefile contained the area data that was used to make the 30% block group area assessment for defining HealthLine access. Ultimately some data was lost in the merger due to geographic definition differences. Any observations that had missing data were removed from the final dataset.

²²U.S. Census Bureau, "LEHD Origin-Destination Employment Statistics Data (2002-2021)," U.S. Census Bureau, Longitudinal-Employer Household Dynamics Program, <https://lehd.ces.census.gov/data/#lodes>, LODES8.1.

²³Steven Manson et al., *IPUMS National Historical Geographic Information System: Version 18.0 2009 American Community Survey: 5-Year Data [2005-2009, Block Groups & Larger Areas]*, Minneapolis, MN, <http://doi.org/10.18128/D050.V18.0>.

²⁴Steven Manson et al., *IPUMS National Historical Geographic Information System: Version 18.0 2013 American Community Survey: 5-Year Data [2009-2013, Block Groups & Larger Areas]*, Minneapolis, MN, <http://doi.org/10.18128/D050.V18.0>.

²⁵Steven Manson et al., *IPUMS National Historical Geographic Information System: Version 15.0 2000 to 2010 Block Group Parts to Block Groups Crosswalk*, Minneapolis, MN, <http://doi.org/10.18128/D050.V15.0>.

²⁶Steven Manson et al., *IPUMS National Historical Geographic Information System: Version 16.0 2010 to 2020 Block Groups to Block Groups Crosswalk*, Minneapolis, MN, <http://doi.org/10.18128/D050.V16.0>.

²⁷Cuyahoga County Open Data, *U.S. Census Blocks: National Geospatial Data Asset - ESRI US Federal Data*, <https://data-cuyahoga.opendata.arcgis.com/datasets/fedmaps::u-s-census-blocks-1/about>.

4.1 LEHD LODES Dataset

The LEHD LODES Dataset contains the key variables of interest. The data is a tabulation of the number of jobs in a given year according to a certain category. All the data in this data set has been normalized on the population totals obtained from the 5-year ACS and the subsequent weighting from the geographic crosswalks. The variables of interest that are used are as follows:

Original Data Definition from LEHD	Data Definition after ACS normalization
Total number of jobs	Percent of total population employed
Total jobs in healthcare and social assistance industries	Job share of healthcare industry
Total jobs in food service and accommodation industry	Job share of food service and accommodation industry
Total jobs in educational services Industry	Job share of educational services industry
Total jobs in scientific, technical and professional services industry	Job share of scientific, technical and professional services industry

Table 1: Data Definitions and Employment Statistics from LEHD LODES RAC data

4.2 5-Year ACS Data

The ACS Data is similarly tabulated as population totals. As such, all demographic categories are normalized on the total population. The normalization process yielded variables for the percent of population that is White, percent of population that is Black, percent of population that is Asian, percent of population that is Non-White Hispanic, percent of population that is Male, percent of population that is Female, percent of population with a high school diploma or equivalent degree, percent of population with some college education but no degree, percent of population with an Associate's degree, percent of population with a Bachelor's degree, percent of population with a Master's degree, the percent of population with a professional advanced degree, and the percent of population with a Doctorate. Furthermore a "Has Degree" variable was created by summing the percentage of the population with at least an Associate's degree.

5 Data Exploration

5.1 Summary Statistics

Table 2 shows the mean values of demographic, employment, and educational attainment data for Census Block Groups with access to the HealthLine BRT and those without access, as well as the p-value from the independent samples t-test for statistical significance. It is clear from the summary statistics that the two groups are not balanced. Residents living with access to the HealthLine BRT are less white, lower income, and less educated as a whole.

The Census Block Groups around the Euclid Corridor are, on average, 46 percentage points more black than those not around the Euclid Corridor. This is significant at the 0.1% level. The only demographic similarities are the population shares of children, and the elderly.

Looking at Employment, it can be seen that the Block Groups of the Euclid Corridor have, on average, a working population percentage that is 8 percentage points less than non-Euclid Corridor Block Groups.

Table 2: Summary Statistics of Cleveland Block Groups

	HealthLine Access		Rest of Cuyahoga Cty.		
	mean	std. dev.	mean	std. dev.	p-value
<i>Demographics</i>					
White	0.169	0.224	0.649	0.341	< 0.001***
Black	0.757	0.302	0.296	0.349	< 0.001***
Asian	0.046	0.106	0.022	0.038	< 0.001***
Non-White Hispanic	0.019	0.035	0.048	0.086	< 0.001***
Male	0.498	0.098	0.473	0.053	< 0.001***
Female	0.502	0.098	0.527	0.053	< 0.001***
Under 18	0.235	0.124	0.236	0.078	0.827
Over 65	0.185	0.113	0.184	0.084	0.930
Average Household Size	2.601	1.133	2.430	0.440	< 0.001***
<i>Employment</i>					
% of pop. employed	0.373	0.143	0.452	0.135	< 0.001***
Job share earning < \$1250/month	0.348	0.079	0.284	0.056	< 0.001***
Job share earning btwn. \$1250 and \$3333/month	0.415	0.074	0.382	0.086	< 0.001***
Job share earning over \$3334/month	0.237	0.127	0.334	0.126	< 0.001***
Job share of health care and social assistance	0.232	0.065	0.179	0.052	< 0.001***
Job share of food service and accommodation	0.096	0.025	0.083	0.021	< 0.001***
<i>Educational Attainment</i>					
No GED	0.163	0.079	0.094	0.072	< 0.001***
HS or GED equiv.	0.205	0.083	0.208	0.083	0.577
Some College	0.134	0.052	0.149	0.051	< 0.001***
Associates Degree	0.030	0.022	0.047	0.027	< 0.001***
Bachelors Degree	0.061	0.062	0.115	0.078	< 0.001***
Masters Degree	0.031	0.032	0.052	0.047	< 0.001***
Adv. Professional Degree	0.013	0.026	0.018	0.029	0.0016**
Doctorate	0.007	0.011	0.008	0.015	0.136

p-values represent difference between groups with and without access

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Furthermore, at the less than \$1250 per month, or the \$15,000 per year income level,²⁸ there is, on average, 6 percentage points more of the workforce earning wages that put them at or just above the poverty threshold. Considering the larger household sizes, on average, of the Euclid Corridor residents this effect may further be exacerbated by those additional costs.

Furthermore, the Euclid Corridor is less educated than the rest of Cuyahoga County. Euclid Corridor Block Groups have, on average, 6 percentage points more residents with no HS diploma or GED. Furthermore, Euclid Corridor Block Groups, also see 5 percentage points, on average, less residents with a Bachelors degree. These educational disparities contribute to the overall economic disparity seen between the Euclid Corridor and the rest of Cuyahoga County.

²⁸ About 150% the single person poverty threshold, or just over 100% of the two person household threshold, and below the three person household threshold.

5.2 Data Visualization

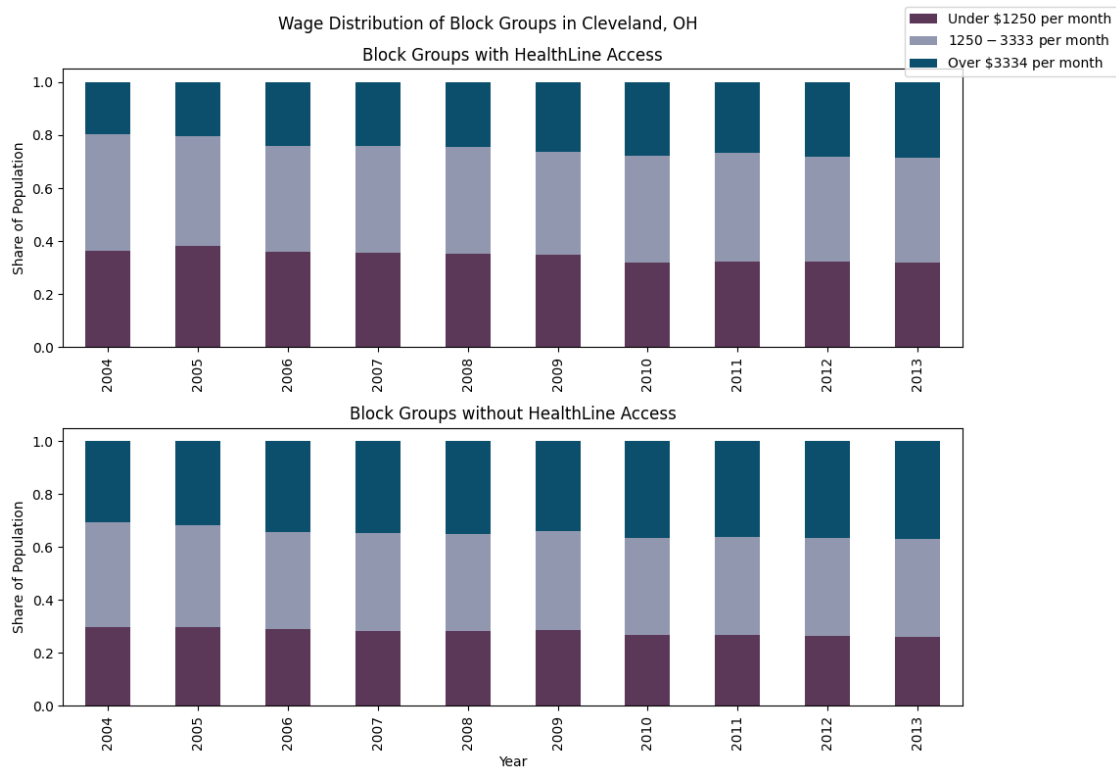


Figure 2: Distribution of monthly income between Block Groups with access to the HealthLine and those without from 2004-2013

Figure 2 shows the distribution of employee wages as they fall into their respective income bins. It can be seen that Census Block Groups not on the Euclid Corridor, a higher proportion of their jobs are in the highest income category. It also appears that the Euclid Corridor block groups see a shrinking of the middle income category that corresponds with increases to the highest income and slight decreases to the lowest income bracket. Overall it appears that there is more steady changes in the control group rather than the Euclid Corridor group. Potentially suggesting less volatility in their job market, and more stable benefits. It is also clear that it appears that the proportion earning between \$15,000 a year and \$39,996 a year is about the same in both groups, or at least is the most similar. With the real differences appearing to be within the below \$15,000 a year group and the over \$39,996 a year group.

Figure 3 shows the three income definitions and overall employment of the Euclid Corridor and non-Euclid Corridor groups both pre and post treatment. This figure also reveals that the Euclid Corridor Census Block Groups have a higher percentage of workers earning between \$15,000 and \$39,996 per year. Furthermore, it confirms that the HealthLine does appear to have a more volatile wage trend, with the middle income group having more variance than the control group. It also shows that there really isn't a major effect on wages because of the HealthLine opening. In fact, it appears that for the most part it stays the same, with the exception of the middle income group in which they see a small spike, but no real held effects compared to the less variant but ultimately similar control group. Looking at the percent employed, we can see that

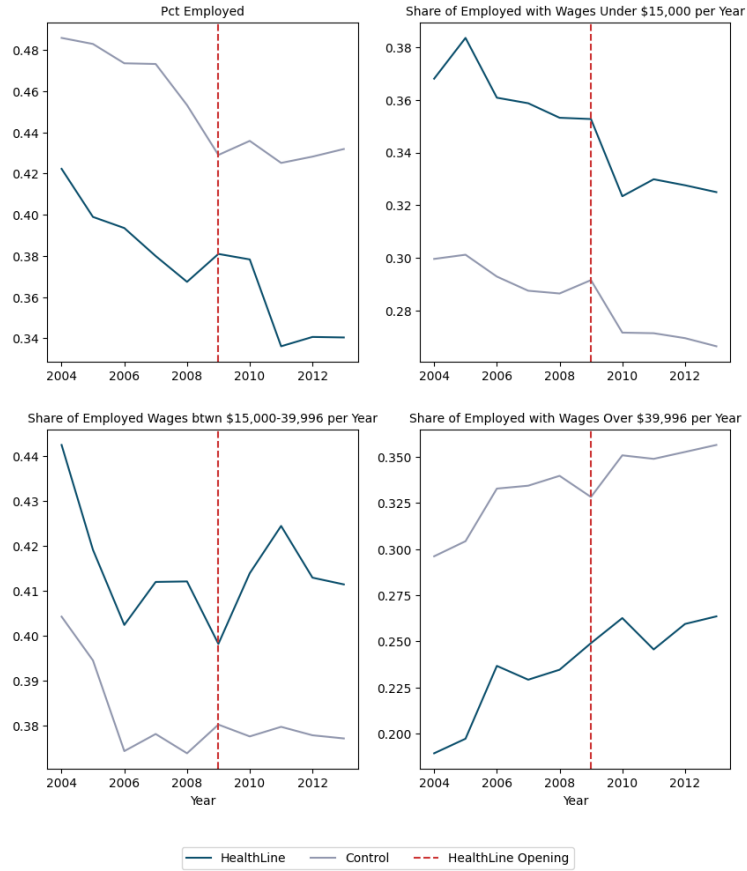


Figure 3: Parallel Trends in pre-treatment income groups and overall employment

there is not any real effect despite the pre-trends. In fact, there is a major decrease in 2011 compared to the rest of Cuyahoga County. It is unclear what is causing this potential downward spike post opening. Overall we can conclude that the effect is basically none for overall employment.

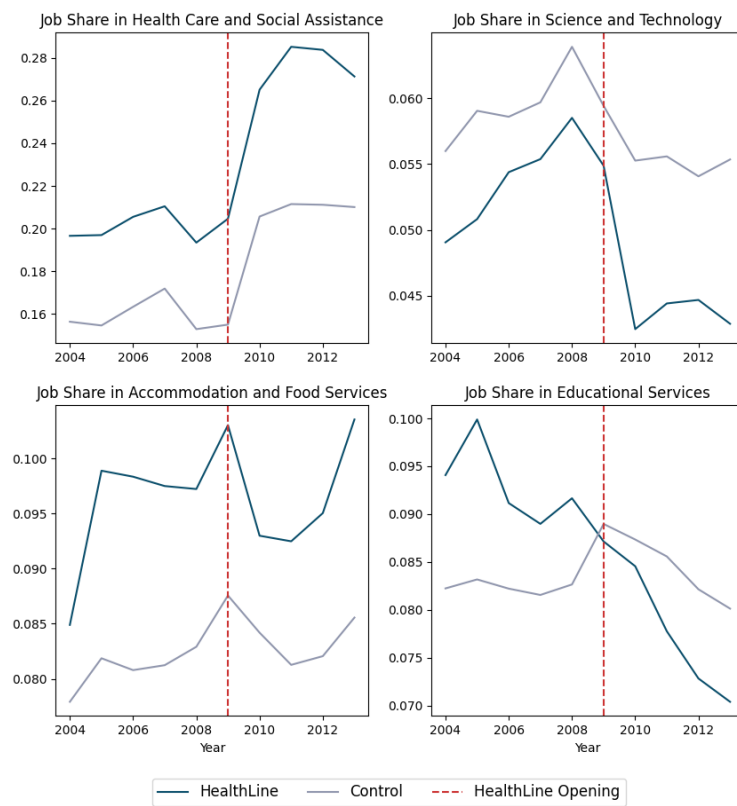


Figure 4: Parallel trends in pre-treatment outcomes

Figure 4 depicts the parallel trends between the treatment–Euclid Corridor–Census Block Groups and the control Census Block Groups, over time, with start of treatment showing with the red line. It shows the effects for the percentage of jobs in health care and social assistance; professional, scientific, and technical services; accommodation and food services; and educational services. There is solid evidence of parallel pre-trends for both health care and social assistance and professional, scientific, and technical services. There is a boom in health care and social assistance for the HealthLine group, with the jobshare of health care and social assistance increasing more than the control group. This implies that Euclid Corridor residents did in fact benefit on the healthcare front with University Hospitals and the Cleveland Clinic investing in the creation and naming of the HealthLine. However that coincided with a sharp decline in science and tech job share for the Euclid Corridor. While both groups saw a decrease, the educational disparity likely made Euclid Corridor residents less desirable in this industry. Finally, for accommodation and food services, we do not observe parallel pre-trends as we see a large positive spike in 2005. This can likely be attributed to anticipatory behavior following the announcement of the HealthLine in 2004. It is likely that businesses began to open up in the accommodation and food services industry in anticipation of the increased economic activity that would follow the opening of the HealthLine, thus the vast majority of new food service jobs would have already been allocated prior to the opening of the HealthLine. There was a spike around 2008 during the opening, but later a decline before a sharper rise than the control group. This implies that post recession, as economic activity picked up, accommodation and food services, likely around the HealthLine, benefited from the additional activity, and that those benefits were passed on the Euclid Corridor residents.

6 Empirical Analysis

6.1 Difference-in-Difference Estimator

To evaluate the effect of the HealthLine BRT opening on overall employment, and job shares of several key industries a difference-in-difference approach was used. For each outcome variable, a 2-way fixed effects Difference in Difference regression, and three Triple Difference-in-Difference—for analysis of interaction terms—were run. There were 5 outcome variables for each of these regressions (percent of total population employed; Job share of healthcare and social assistance industry; Job share of professional, scientific and technical services industry; Job share of education industry; Job share of the food service and accommodation industry). The equations of the regressions are as seen below:

1. Two-Way Fixed Effects Difference in Differences:

$$Y_{itj} = \beta_0 + \alpha_t + \alpha_i + \beta_3 \text{HealthLine} \times \text{After}_{it} + \mu_{it} + \epsilon_{it}$$

2. Triple Difference-in-Differences:

$$Y_{itj} = \beta_0 + \beta_1 \text{HealthLine}_{it} + \beta_2 \text{After}_{it} + \beta_3 \text{HealthLine} \times \text{After}_{it} + \beta_4 \text{Interaction} \times \text{After}_{it} \\ + \beta_5 \text{Interaction} \times \text{HealthLine}_{it} + \beta_6 \text{Interaction} \times \text{HealthLine} \times \text{After}_{it} + \beta_\mu \mu_{it} + \epsilon_{it}$$

Where:

i : Census Block Group (2020 Census Designation)

j : Outcome variable

t : Year (2004-2013)

Y : Outcome variable (Listed above)

μ : Covariate Controls (listed below)

ϵ : Error

α_t : Time Fixed Effects (Year)

α_i : Location Fixed Effects (Census Block Group)

The covariate controls used were Percent of Census Block Group population that is Black, Percent of Census Block that has not obtained a GED, Percent of Census Block Group population that has at least an Associates Degree, and Median Home Value of the Census Block Group.

And the interaction terms are defined as follows:

1. Minority → Any Census Block Group with less than 20% of the population White, (Cleveland as a whole is 38% White).
2. No GED → Any Census Block Group with over 22% of the population not having a GED or equivalent (Cleveland as a whole has 18% of the population with no GED).
3. Degree → Any Census Block Group with over 30% of the population having at least an Associates Degree. (The Cuyahoga County overall mean is 23%).

6.1.1 Difference-in-Differences Results

Table 3 shows the results from running the two-way fixed effects difference-in-differences regression on our outcome variables. We can see that food service and % employed do not yield statistically significant results. Healthcare, Education, and Science and Technology all have statistically significant results. Overall employment does not see a significant effect, although there is a negative, non-significant, association with the opening of the HealthLine. However, due to the Great Recession of 2008, it is difficult to tell whether this negative effect is a true effect. It is likely due to recession hitting the Euclid Corridor communities harder than other communities in Cleveland, although not significantly so. Food service and accommodation saw essentially 0 effect, this is likely because of the spike in 2005 seen in Figure 4. This anticipatory action

Table 3: Two-Way Fixed Effects Difference-in-Differences Regressions

	(1)	(2)	(3)	(4)	(5)
	% employed	Healthcare	Food Service	Education	Science & Tech
Healthline $\times$ After	-0.0117 (0.0114)	0.0324*** (0.00414)	0.0000963 (0.00244)	-0.0125*** (0.00240)	-0.0101*** (0.00183)
% Black	-0.0622** (0.0209)	-0.00456 (0.00596)	0.00355 (0.00372)	-0.000811 (0.00370)	-0.00336 (0.00287)
% Has Degree	0.357*** (0.0293)	-0.0452*** (0.00793)	-0.0145** (0.00464)	0.0318*** (0.00499)	0.0326*** (0.00427)
% No GED	0.184*** (0.0415)	-0.0132 (0.0104)	0.00902 (0.00627)	0.00764 (0.00610)	0.00242 (0.00480)
Med. Home Val.	-0.00000264*** (2.84e-08)	-1.49e-08* (6.91e-09)	-9.36e-10 (3.49e-09)	-3.37e-09 (4.10e-09)	2.90e-08*** (3.71e-09)
Constant	0.405*** (0.0121)	0.196*** (0.00330)	0.0846*** (0.00198)	0.0764*** (0.00201)	0.0465*** (0.00165)
Two-Way Fixed Effects?	Yes	Yes	Yes	Yes	Yes
N	11066	11066	11066	11066	11066
R ²	0.597	0.785	0.526	0.746	0.669

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

made it so that there was no major change following 2008, but it can be assumed that the spike in 2005 was likely a result of the HealthLine indirectly.

The opening of the HealthLine causes a 3 percentage point increase, significant at the 0.1 % level, in the healthcare job share. This implies that the additional healthcare jobs that Nelson et al. found likely did go to Euclid Corridor residents at a higher rate than non Euclid Corridor residents. This shows the efficacy of investing in public transportation infrastructure. The key industry of the area sees increased jobs, and those on the line see the increased opportunities going to them. Education and Science and Technology industries, however, experience a statistically significant decrease in job share following the opening of the HealthLine. This is an interesting result to see. Considering the HealthLine being attributed with sparking the Health-Tech corridor, the lack of expansion of the Science and Technology sectors for resident Block groups seems counter-intuitive. However, when considering Table 2, we see that the Euclid Corridor is significantly less educated and less White. This sparks the question of whether there is potentially a racial component to this decrease, or whether this is a case of qualification differences as a result of the educational disparities experienced by Euclid Corridor residents.

Furthermore, the % has degree and HealthLine effect (HealthLine $\times$ After) see opposite signs for each outcome variable. The % has degree coefficient is also statistically significant for each outcome variable. This implies that education is a major driver of employment outcomes, and appears to run opposite to the HealthLine effect.

The education and HealthLine trends, as well as the decreases in Science and Technology as well as the Education sectors prompted a further study of how education and race interact with the HealthLine effect. To study the intersectionality of public transit access with race and education we used a Triple Difference-

in-Differences regression model—defined previously in section 6.3—in which dummy variables are used to explore certain smaller groups. The thresholds were chosen based on the Cleveland and Euclid Corridor averages, in order to create groups with enough data points to estimate the intersectional effects.

6.2 Triple Difference-in-Differences Estimator

6.2.1 Percent Employed

Table 4: TWFE Interaction Term Triple Diff-in-Diff on % Pop. Employed

	(1) Minority Interaction	(2) No GED Interaction	(3) >Assoc. Deg. Interaction
Healthline $\times$ After	0.0726** (0.0266)	-0.00754 (0.0140)	-0.0234* (0.0108)
> 80% Minority	-0.00817 (0.00925)	-0.0698*** (0.00925)	-0.0673*** (0.00907)
> 30% Has Degree	0.0173** (0.00570)	0.0224*** (0.00573)	0.00667 (0.00629)
> 22% No GED	-0.00326 (0.00642)	-0.00775 (0.00752)	-0.00566 (0.00651)
HealthLine $\times$ Interaction	-0.190*** (0.0220)	0.0276 (0.0225)	-0.127** (0.0415)
After $\times$ Interaction	-0.0614*** (0.00466)	0.00858 (0.00750)	0.0278*** (0.00424)
After $\times$ HealthLine $\times$ Interaction	-0.0613* (0.0299)	-0.00892 (0.0502)	0.0990* (0.0440)
Constant	0.456*** (0.00261)	0.457*** (0.00277)	0.457*** (0.00273)
N	11066	11066	11066
R^2	0.596	0.587	0.589

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4 shows the Triple Difference-in-Difference regressions on % of total population employed. Firstly, when looking at our minority interaction, defined as any Census Block Group with over 80% of its population being non-White, we see that the opening of the HealthLine led to a 6.13 percentage point decrease, statistically significant at the 5% level, to overall employment. This is contradictory to our previous results in Table 3 which saw a statistically insignificant decrease. Instead here we see that for Block Groups with significant minority populations there were employment losses. This is again likely due to the Great Recession hurting minorities in Cleveland, as seen by the After $\times$ Interaction coefficient which is significant at the 0.1% level. look at the overall HealthLine effect for these block groups we can see, there is a 1 percentage point increase. This implies that while there was a slight decrease in jobs, for these significantly minority communities there was actually a slight increase in jobs when compared to other significantly minority communities not in the Euclid corridor. This implies that there is a shrinkage of the

racial employment disparity to some extent in Euclid Corridor communities as a result of the HealthLine. But while there is some decrease when compared to non Euclid Corridor groups, we still see an overall disparity trend within the Euclid Corridor, where Block groups with less minority population actually see improvements to employment, while these over 80% minority block groups see a decrease. This tells us that while the addition of public transportation does improve minority communities employment outcomes when compared to those without access they do not receive the same level of benefit as block groups with less minorities.

We do not find statistically significant results for lower educated groups. However, for Census Block Groups with over 30% of its population having attained at least an Associates Degree, the HealthLine opening led to a 9.9 percentage point increase in employment—statistically significant at the 5% level. This implies that there is indeed an educational component to employment. We can see that that compared to non Euclid Corridor groups with the same educational attainment there is a roughly 7.6 percentage point increase in employment as a result of the HealthLine. We also see that educated block groups do better than non educated groups on the HealthLine. This result can have significant policy implications when considering the clear positive effect that more educated block groups experience as a result of the increased access to public transit.

6.2.2 Healthcare Industry

Table 5: TWFE Interaction Term Triple Diff-in-Diff on Healthcare Job Share

	(1) Minority Interaction	(2) No GED Interaction	(3) >Assoc. Deg. Interaction
HealthLine $\times$ After	0.0295** (0.00944)	0.0370*** (0.00497)	0.0311*** (0.00429)
> 80% Minority	-0.0121** (0.00368)	0.0205*** (0.00405)	0.0207*** (0.00398)
> 30% Has Degree	-0.00619*** (0.00131)	-0.00976*** (0.00135)	-0.00131 (0.00150)
> 22% No GED	-0.000422 (0.00194)	-0.00502* (0.00228)	0.00137 (0.00201)
HealthLine $\times$ Interaction	0.0341 (0.0199)	0.00962 (0.00943)	0.0269 (0.0228)
After $\times$ Interaction	0.0492*** (0.00137)	0.0122*** (0.00250)	-0.0156*** (0.00115)
After $\times$ HealthLine $\times$ Interaction	-0.0332** (0.0104)	-0.0469*** (0.0138)	-0.00465 (0.0127)
Constant	0.179*** (0.000915)	0.179*** (0.000949)	0.179*** (0.000940)
N	11066	11066	11066
R^2	0.815	0.787	0.790

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

We find that while the Euclid Corridor saw a 3.24 percentage point increase in Healthcare job share, significant at the 0.1% level (seen in Table 3), Block groups that were over 80% minority actually saw a 3.32% additional decrease, statistically significant at the 1% level, essentially removing any positive effect seen by the Euclid Corridor. This brings a massive caveat to the associated boost to healthcare jobs. It further emphasizes the racial disparities that Euclid Corridor residents endure. Block groups with over 80% minorities see essentially no effect compared to non-Euclid Corridor groups.

We also see that there is an education component in the healthcare industry. Healthcare job share sees an additional decrease of 4.69 percentage points, statistically significant at the 0.1% level, for block groups with over 22% of its residents without a GED. As for the more educated groups, groups with over 30% of its population having attained at least an Associates Degree, there is no significant additional effect. So while groups with more educational attainment do not see any major additional benefits beyond the overall HealthLine boost, groups with larger percentages of people without a GED see an additional decrease in Healthcare jobs. This result offers a potential reason for the racial disparities seen as the shift in the minority dummy coefficient from negative to positive implies that the no GED dummy variable is absorbing a lot of that negative result. This implies that the racial disparity seen in healthcare job share may be explained away by the education gap, and the failure to meet job qualifications as a result of the lack of a GED.

6.2.3 Food Service & Accommodation Industry

Table 6: TWFE Interaction Term Triple Diff-in-Diff on Food Service & Accommodation Job Share

	(1) Minority Interaction	(2) No GED Interaction	(3) >Assoc. Deg. Interaction
HealthLine $\times$ After	-0.00152 (0.00626)	-0.00405 (0.00253)	0.00239 (0.00270)
> 80% Minority	-0.00381 (0.00216)	-0.00473* (0.00207)	-0.00490* (0.00204)
> 30% Has Degree	-0.000479 (0.000816)	-0.000103 (0.000813)	0.00296*** (0.000882)
> 22% No GED	0.00303* (0.00132)	0.0000579 (0.00155)	0.00353** (0.00131)
HealthLine $\times$ Interaction	0.00136 (0.00858)	-0.000982 (0.00553)	-0.00120 (0.00729)
After $\times$ Interaction	-0.00167 (0.000871)	0.00643*** (0.00176)	-0.00546*** (0.000628)
After $\times$ HealthLine $\times$ Interaction	0.00413 (0.00692)	0.0552*** (0.0116)	-0.0184** (0.00603)
Constant	0.0838*** (0.000514)	0.0837*** (0.000510)	0.0837*** (0.000507)
N	11066	11066	11066
R^2	0.815	0.787	0.790

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 3 shows that there is no statistically significant effect of the HealthLine opening on food service job share. However, there is an interesting relationship shown in Figure 4. Figure 4 shows a large spike in job share in the years leading to the opening of the HealthLine. It is possible that there was a future expectations effect leading up to the HealthLine opening. It would make sense, that upon learning of the new major transit line opening on the Euclid Corridor, businesses would start opening in and around the HealthLine hoping to capture the increased economic activity that the additional commuters would bring.

Furthermore Table 6 shows interesting implications for neighborhoods with less educational attainment. While there are no significant additional effects seen for block groups with significant minority populations. However, for Euclid Corridor blocks with over 22% of its population without a GED, there is an additional 5.52 percentage point increase in food service and accommodation job share. This implies that the addition of public transportation increased the food service and accommodation job share significantly for block groups with more lower educated persons.

6.2.4 Education Industry

Table 7: TWFE Interaction Term Triple Diff-in-Diff on Education Job Share

	(1) Minority Interaction	(2) No GED Interaction	(3) >Assoc. Deg. Interaction
Healthline $\times$ After	0.00659 (0.00589)	-0.0112*** (0.00297)	-0.0146*** (0.00229)
> 80% Minority	0.00785*** (0.00215)	-0.00275 (0.00203)	-0.00265 (0.00198)
> 30% Has Degree	0.00517*** (0.000914)	0.00597*** (0.000920)	-0.000591 (0.00102)
> 22% No GED	0.000177 (0.00115)	0.00429*** (0.00126)	-0.000562 (0.00113)
HealthLine $\times$ Interaction	-0.00199 (0.00727)	0.00501 (0.00533)	-0.0127 (0.0146)
After $\times$ Interaction	-0.0163*** (0.000808)	-0.0105*** (0.00135)	0.0119*** (0.000725)
After $\times$ HealthLine $\times$ Interaction	-0.0134* (0.00653)	0.0112 (0.00838)	0.0288** (0.00876)
Constant	0.0823*** (0.000526)	0.0825*** (0.000531)	0.0827*** (0.000524)
N	11066	11066	11066
R^2	0.757	0.747	0.755

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

While the HealthLine was associated with a decrease in education jobs, we again see an interesting education and race effect in Table 7. We see that block groups with over 80% of its population being minorities see a 1.34 percentage point additional decrease, significant at the 5% level. Implying that there is a negative effect for racial minorities on the HealthLine when it comes to education job share.

As for the educational gap, we see a major benefit for more educated groups in the Euclid Corridor, with an approximately 4.2 percentage point increase in job share for more educated groups when compared to other groups in the Euclid Corridor. When compared to non Euclid Corridor block groups, we see that there is a net 1 percentage point increase in education job share for the educated groups on the HealthLine compared to not on the HealthLine. This implies again that for educated groups, the additional public transportation expansion has positive effects on those who can access it compared to similar demographic groups who do not gain access to these public transit options.

6.2.5 Science and Technology Industry

Table 8: TWFE Interaction Term Triple Diff-in-Diff on Science and Tech Job Share

	(1) Minority Interaction	(2) No GED Interaction	(3) >Assoc. Deg. Interaction
HealthLine $\times$ After	0.00396 (0.00502)	-0.00865*** (0.00233)	-0.0114*** (0.00148)
> 80% Minority	0.00581*** (0.00150)	-0.00911*** (0.00160)	-0.00880*** (0.00153)
> 30% Has Degree	0.00247** (0.000838)	0.00402*** (0.000857)	-0.00306*** (0.000923)
> 22% No GED	0.000678 (0.000834)	0.00307** (0.00104)	-0.000120 (0.000885)
HealthLine $\times$ Interaction	-0.00501 (0.00595)	0.00467 (0.00372)	0.0202* (0.00970)
After $\times$ Interaction	-0.0229*** (0.000566)	-0.00648*** (0.00115)	0.0120*** (0.000593)
After $\times$ HealthLine $\times$ Interaction	-0.00223 (0.00535)	-0.00154 (0.00532)	0.0213*** (0.00639)
Constant	0.0579*** (0.000409)	0.0581*** (0.000432)	0.0582*** (0.000422)
N	11066	11066	11066
R^2	0.705	0.667	0.682

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 8 shows that there is a statistically significant (at the 0.001 level) effect in which the opening of the HealthLine is shown to cause a 1.2 percentage point decrease in science and tech job share. However most notably when looking at the Degree interaction table (column 5 in Table 8), it can be seen that there is a statistically significant positive relationship for Block Groups with a more educated population around the HealthLine and science and tech jobs. This suggests that for residents with higher educational attainment, the HealthLine led to increased opportunities in the tech and science jobs that the HealthLine helped spur on by connecting the Health-Tech Corridor. This suggests that this additional proximity was of a major benefit to degree holding, and thus highly qualified, job seekers. We did not find significant results for minority or lower educated block groups.

7 Shortcomings

One limitation of our research regards the data of our control variables used in the regression analysis. While our primary outcome variable (employment statistics) was sourced from the LODES dataset, which offers data at the census block level and is measured yearly, finding similar data for our controls was not possible. Thus, we chose to utilize the American Community Survey's (ACS) 5 year data. The data is a 5 year tabulation of population statistics at the Census block group level. However, the start of the 5-year ACS data collection started with a 2004-2009 collection, and then continued yearly after. This is an issue for our research because 2009 is the first year of treatment – as the HealthLine opened up in late 2008. Because of this there is some overlap of demographic statistics used in our controls. Ultimately, they serve as a control for neighborhood demographics on a whole, and the one year of potential overlap is unlikely to have a large effect on our analysis as demographics do not often rapidly change over a 1 year period.

Another limitation of our research pertains to the use of % Pop Employed as our outcome variable. While this metric accurately reflects the number of individuals employed at a specific point in time, it does not directly measure the employment rate. This discrepancy arises from the absence of data on the working population at the block group level, resulting in our outcome variable being influenced by factors beyond employment rates, such as age demographics (e.g., children and seniors). Despite examining age-related variables for correlation with HealthLine access and finding no statistical significance, this limitation warrants acknowledgment, and it is noteworthy to mention that using employment rates as the outcome variable would yield more robust and easily interpretable results.

The analysis of income effects, is similarly limited by the LODES data set. The LODES data set reports income in three bins (under \$15,000 a year, between \$15,000 and \$39,996 a year, and over \$39,996 a year). Because of the lack of fineness of this data it was not possible to make out real income effects other than the number of jobs that fall within these bins. Because of this, income effects were left out of the empirical analysis portion of the paper.

Ideally, it would have been possible to look at employment by race. Unfortunately, LODES does not provide job tabulations by employee race prior to 2009. Because of the lack of data in the pre-treatment period it was not possible to use employment by race in this study. Instead, total employment and racial demographics of the Census Block Groups were used instead. This is likely a good proxy, as if Block Groups with a higher Black population are seeing effects on jobs, it is reasonable to expect that there is a similar effect on Black employment rate, etc. However, because of this we could not look into the intersectional nature of the educational and racial gaps in employment. Since we could not look at racial job counts and job counts by educational attainment this created a blind spot in our study for these intersectional effects that we know exist.

8 Discussion and Conclusion

While this paper does not find significant overall employment boosts as a result of the opening of the HealthLine for resident Block Groups, it does find significant increase in Healthcare job share. This implies that for key industries in the area (which healthcare is to the Euclid Corridor), there is increased employment.

This does indicate that, broadly, public transportation access expansion does lead to increased opportunity, especially in the industries that expand as a result of the expansion. However, the triple difference-in-difference results raise important questions about racial and educational effects for policy makers to consider in the future.

Through triple difference-in-differences regression, it is found that the HealthLine opening led to increased employment for more educated Block groups compared to similarly educated non-HealthLine groups, implying that increasing access to public transportation is particularly effective for those who are educated. This indicates that there is an educational component to employment disparity. Access is not the only component. Future policymakers should look into how public transit development can be implemented in conjunction with increased educational access for lower educated neighborhoods as a tool for economic revitalization, and increased social mobility.

This study also finds that there is a potential increase in job market robustness for block groups with significant minority populations. Importantly, racial disparities continue to perpetuate within the areas that gain additional access. So while these communities may be slightly more recession-resistant compared to similar block groups elsewhere, they do not see the same level of benefit as block groups with smaller minority populations see. This indicates that future policymakers need to be mindful about how they can work to implement public transit development in ways that help alleviate racial disparities. Unfortunately, due to data limitations, I was unable to look at how much these racial disparities in employment were connected to racial disparities in educational attainment, although I do theorize that this is a major component.

A major blind spot in this research is the lack of precise wage data. Because of this wage effects in the Euclid Corridor cannot be studied, only changes in employment. In the future I would like to see how wages are affected by public transit expansion. This would help understand whether these increases in employment coincide with decreased wages overall, or if public transit access on its own can play a major role in increasing the wealth of nearby communities.

In the future, we believe that research should be conducted looking into the intersection of educational and transit access. For block groups with a higher percentage of people with at least an Associates degree, they saw improved employment outcomes compared to both groups with less educational attainment, but also compared to similar groups that did not gain access to the HealthLine. It is with this result in mind, Table 4, that we believe looking into public transit access combined with expanded community college access is important.

Ultimately, the findings show that public transit expansion has net positive effects on resident communities, with additional benefits going to those who were higher educated and thus more qualified. As a result of this research, we believe that public transit expansion, specifically connecting communities to major employment centers, can provide a major boost to the residents who are impacted. As Cleveland continues to invest in public transit, we believe that if effectively done, opportunities can be distributed to communities who in the past had been left behind, like the Euclid Corridor.

References

- Allard, Sam. *RTA gets federal grant for Lorain Corridor*, Apr. 2024. <https://www.axios.com/local/cleveland/2024/04/09/rta-federal-grant-lorain-corridor-brt>.
- Brown Announces \$264 Million for Ohio's Public Transit Systems — U.S. Senator Sherrod Brown of Ohio*. <https://www.brown.senate.gov/newsroom/press/release/sherrod-brown-announces-264-million-for-ohios-public-transit-systems-fy2023>.
- Clinic, UH pay to name Euclid Corridor buses - cleveland.com*. https://www.cleveland.com/medical/2008/02/clinic_uh_pay_to_name.html.
- Credit, Kevin N. “Economic development for the 21st century: how proximity to transit and walkability influence business creation and performance”. PhD thesis, Michigan State University, 2018. <https://doi.org/doi:10.25335/bvec-qy72>.
- Data, Cuyahoga County Open. *U.S. Census Blocks: National Geospatial Data Asset - ESRI US Federal Data*. <https://data-cuyahoga.opendata.arcgis.com/datasets/fedmaps::u-s-census-blocks-1/about>.
- Epstein, Jeff. “Cleveland Health-tech corridor”. *Economic Development Journal* 15, no. 2 (2016): 7–13. ISSN: 1539-1922. <https://web.p.ebscohost.com/ehost/pdfviewer/pdfviewer?vid=2&sid=80851b1c-cbfe-4b7c-b5f0-2f2906505ecd@redis>.
- Euclid Avenue Healthline Bus Rapid Transit*. <https://www.sasaki.com/projects/euclid-avenue-healthline-bus-rapid-transit/>.
- Manson, Steven, et al. *IPUMS National Historical Geographic Information System: Version 15.0 2000 to 2010 Block Group Parts to Block Groups Crosswalk*. Minneapolis, MN. <http://doi.org/10.18128/D050.V15.0>.
- . *IPUMS National Historical Geographic Information System: Version 16.0 2010 to 2020 Block Groups to Block Groups Crosswalk*. Minneapolis, MN. <http://doi.org/10.18128/D050.V16.0>.
- . *IPUMS National Historical Geographic Information System: Version 18.0 2009 American Community Survey: 5-Year Data [2005-2009, Block Groups & Larger Areas]*. Minneapolis, MN. <http://doi.org/10.18128/D050.V18.0>.
- . *IPUMS National Historical Geographic Information System: Version 18.0 2013 American Community Survey: 5-Year Data [2009-2013, Block Groups & Larger Areas]*. Minneapolis, MN. <http://doi.org/10.18128/D050.V18.0>.
- Nelson, Arthur C., Keuntae Kim, and Joanna P. Ganning. “Bus Rapid Transit and Economic Development: A Quasi-Experimental Treatment and Control Analysis”, 23. Washington DC, United States: University of Arizona: College of Architecture, Planning, / Landscape Architecture, University of Utah: Department of City / Metropolitan Planning, Cleveland State University, Nov. 2016. https://ppms.trec.pdx.edu/media/project_files/Nelson__Kim_-_Bus_Rapid_Transit_and_Economic_Development.pdf.

- Nelson, Arthur C., et al. "Bus Rapid Transit and Economic Development: Case Study of the Eugene-Springfield BRT System". *Journal of Public Transportation* 16, no. 3 (July 2013): 41–57. <https://doi.org/http://doi.org/10.5038/2375-0901.16.3.3>.
- Oct. 24, 2008: *Healthline opens, pumping new life into Cleveland*, Oct. 2008. <https://www.riderta.com/news/oct-24-2008-healthline-opens-pumping-new-life-cleveland>.
- Pasha, Obed, et al. "Transit effects on poverty, employment, and rent in Cuyahoga County, Ohio". *Transport Policy* 88 (Mar. 2020): 33–41. <https://doi.org/https://doi.org/10.1016/j.tranpol.2020.01.013>.
- Piiparinen, Richey, and Jim Russell. "Center for Population Dynamics Quarterly Brief January 2017: Transportation's Role in the Economic Restructuring of Cleveland". *All Maxine Goodman Levin School of Urban Affairs Publications*, Jan. 2017. https://engagedscholarship.csuohio.edu/urban_facpub/1427/.
- Segall, Grant. *RTA says HealthLine had 10-year payback of \$9.5 billion, 'woke up' Euclid Corridor*. news, Nov. 2018. <https://www.cleveland.com/news/erry-2018/11/149927818e3851/rta-says-healthline-had-10year.html>.
- Yi, Chang. "Impact of Public Transit on Employment Status: Disaggregate Analysis of Houston, Texas". *Transportation Research Record Journal of the Transportation Research Board* 1986, no. 1 (Jan. 2006): 137–144. <https://doi.org/http://dx.doi.org/10.1177/0361198106198600117>.